

Program No: 1

Date: 29/06/2026

Program Title: Create an app that displays name and email id.

MainActivity.java:

```
package com.example.labcycle;
import android.annotation.SuppressLint;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.view.View;
import android.widget.EditText;
import android.widget.Button;
import android.widget.TextView;

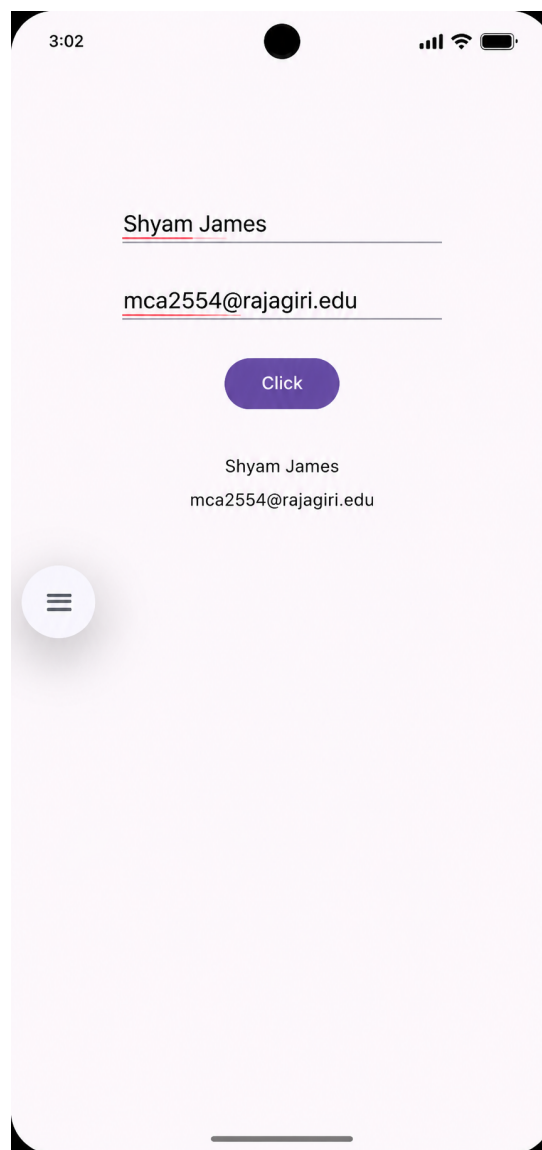
public class MainActivity extends AppCompatActivity {
    EditText name,email;
    Button btn;
    TextView t1,t2;
    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        name = findViewById(R.id.nametxt);
        email = findViewById(R.id.emailTxt);
        btn = findViewById(R.id.btnclick);
        t1 = findViewById(R.id.txtView1);
        t2 = findViewById(R.id.txtView2);
        btn.setOnClickListener(new View.OnClickListener(){
            @Override
            public void onClick(View view) {
                String n1 = name.getText().toString();
                String n2 = email.getText().toString();
                t1.setText(n1);
                t2.setText(n2);
            }
        });
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <EditText
        android:id="@+id/nametxt"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:hint="Name"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="150dp"/>
    <EditText
        android:id="@+id/emailTxt"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:hint="Email"
        app:layout_constraintTop_toBottomOf="@id/nametxt"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp"/>
    <Button
        android:id="@+id/btnclick"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click"
        app:layout_constraintTop_toBottomOf="@id/emailTxt"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="20dp"/>
    <TextView
        android:id="@+id/txtView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name"
        app:layout_constraintTop_toBottomOf="@id/btnclick"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="30dp"/>
```

```
<TextView
    android:id="@+id/txtView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Email"
    app:layout_constraintTop_toBottomOf="@id/txtView1"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="10dp"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 2

Date: 29/06/2026

Program Title: Create an app that takes marks of any three subjects from the user and displays its sum and average.

MainActivity.java:

```
package com.example.labcycle;

import android.annotation.SuppressLint;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.view.View;
import android.widget.EditText;
import android.widget.Button;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
    EditText m1, m2, m3;
    Button btn;
    TextView t1, t2;

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        m1 = findViewById(R.id.mark1);
        m2 = findViewById(R.id.mark2);
        m3 = findViewById(R.id.mark3);
        btn = findViewById(R.id.btncal);
        t1 = findViewById(R.id.textView4);
        t2 = findViewById(R.id.textView5);

        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                int n1 = Integer.parseInt(m1.getText().toString());
                int n2 = Integer.parseInt(m2.getText().toString());
```



```

        int n3 = Integer.parseInt(m3.getText().toString());
        int sums = n1 + n2 + n3;
        int avg = sums / 3;
        t1.setText("Sum is " + sums);
        t2.setText("Average is " + avg);
    }
});
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/mark1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="92dp"
        android:layout_marginBottom="43dp"
        android:ems="10"
        android:inputType="number"
        android:text="mark1"
        app:layout_constraintBottom_toTopOf="@+id/mark2"
        app:layout_constraintEnd_toEndOf="parent" />

    <EditText
        android:id="@+id/mark2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="91dp"
        android:layout_marginBottom="40dp"
        android:ems="10"
        android:inputType="number"
        android:text="mark2"
        app:layout_constraintBottom_toTopOf="@+id/mark3"
        app:layout_constraintEnd_toEndOf="parent" />

    <EditText

```

```

        android:id="@+id/mark3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="85dp"
        android:layout_marginBottom="48dp"
        android:ems="10"
        android:inputType="numberSigned"
        android:text="mark3"
        app:layout_constraintBottom_toTopOf="@+id/btnncal"
        app:layout_constraintEnd_toEndOf="parent" />

<Button
    android:id="@+id/btnncal"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="182dp"
    android:layout_marginBottom="49dp"
    android:text="Button"
    app:layout_constraintBottom_toTopOf="@+id/textView4"
    app:layout_constraintEnd_toEndOf="parent" />

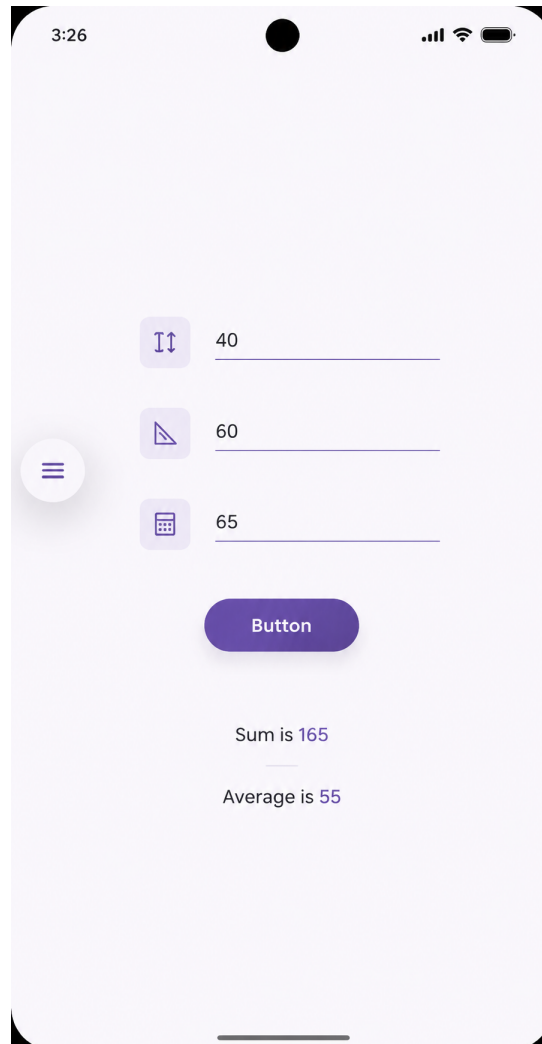
<TextView
    android:id="@+id/textView4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="202dp"
    android:layout_marginBottom="160dp"
    android:text="TextView"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

<TextView
    android:id="@+id/textView5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="200dp"
    android:layout_marginBottom="120dp"
    android:text="TextView"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program No: 3

Date: 29/06/2026

Program Title: Create suitable apps to design the different layouts of the android-studio.

MainActivity.java:

```
package com.example.labcycle;

import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    Button button1, button2, button3;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        button1 = findViewById(R.id.button1);
        button2 = findViewById(R.id.button2);
        button3 = findViewById(R.id.button3);

        button1.setOnClickListener(v ->
            Toast.makeText(this, "LinearLayout Button 1",
                Toast.LENGTH_SHORT).show());
        button2.setOnClickListener(v ->
            Toast.makeText(this, "LinearLayout Button 2",
                Toast.LENGTH_SHORT).show());
        button3.setOnClickListener(v ->
            Toast.makeText(this, "RelativeLayout Button",
                Toast.LENGTH_SHORT).show());
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
```

```

        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

<!-- LinearLayout -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Linear Layout"
    android:textStyle="bold"
    android:textSize="18sp"/>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="20dp">

    <Button
        android:id="@+id/button1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Button 1"/>

    <Button
        android:id="@+id/button2"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Button 2"/>
</LinearLayout>

<!-- RelativeLayout -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Relative Layout"
    android:textStyle="bold"
    android:textSize="18sp"/>

<RelativeLayout
    android:layout_width="match_parent"

```

```

        android:layout_height="120dp"
        android:background="#DDDDDD"
        android:layout_marginBottom="20dp">

        <TextView
            android:id="@+id/text1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Top Left"/>

        <Button
            android:id="@+id/button3"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Bottom Right"
            android:layout_alignParentBottom="true"
            android:layout_alignParentEnd="true"/>
    </RelativeLayout>

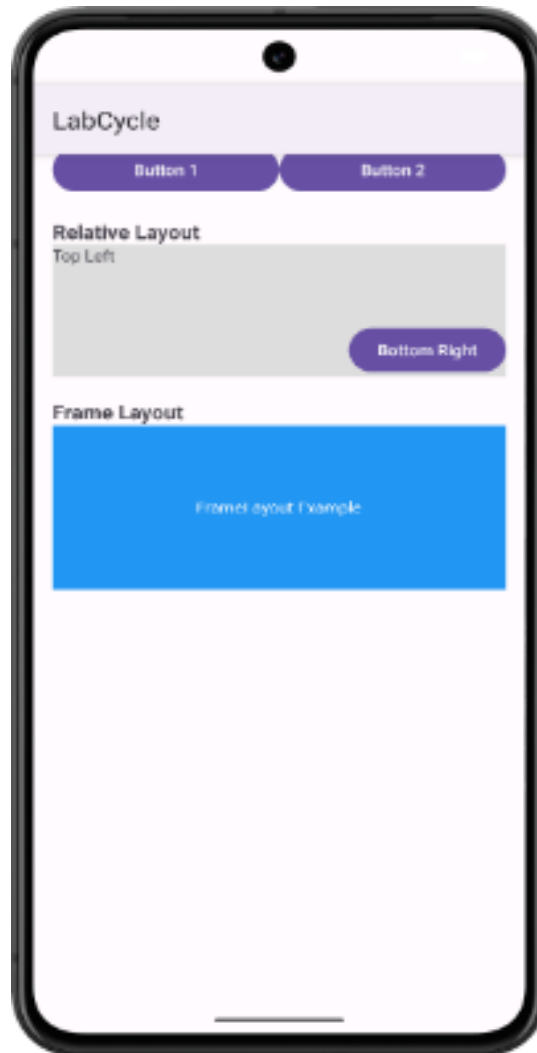
    <!-- FrameLayout -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Frame Layout"
        android:textStyle="bold"
        android:textSize="18sp"/>

    <FrameLayout
        android:layout_width="match_parent"
        android:layout_height="150dp"
        android:background="#2196F3">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="FrameLayout Example"
            android:textColor="#FFFFFF"
            android:layout_gravity="center"/>
    </FrameLayout>
</LinearLayout>
</ScrollView>

```

Output:



Program No: 4

Date: 29/06/2026

Program Title: Develop a simple calculator app.

MainActivity.java:

```
package com.example.pg04.calculator;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.widget.TextView;
import android.widget.Button;
import android.widget.EditText;

public class MainActivity extends AppCompatActivity {
    TextView result;
    Button ad, sub, div, mult;
    EditText no1, no2;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        result = findViewById(R.id.result);
        ad = findViewById(R.id.ad);
        sub = findViewById(R.id.sub);
        div = findViewById(R.id.div);
        mult = findViewById(R.id.mult);
        no1 = findViewById(R.id.no1);
        no2 = findViewById(R.id.no2);

        ad.setOnClickListener(v -> {
            int a = Integer.parseInt(no1.getText().toString());
            int b = Integer.parseInt(no2.getText().toString());
            int res = a + b;
            result.setText(res+"");
        });

        sub.setOnClickListener(v -> {
            int a = Integer.parseInt(no1.getText().toString());
```



```

        int b = Integer.parseInt(no2.getText().toString());
        int res = a - b;
        result.setText(res+"");
    });

    mult.setOnClickListener(v -> {
        int a = Integer.parseInt(no1.getText().toString());
        int b = Integer.parseInt(no2.getText().toString());
        int res = a * b;
        result.setText(res+"");

    });

    div.setOnClickListener(v -> {
        int a = Integer.parseInt(no1.getText().toString());
        int b = Integer.parseInt(no2.getText().toString());
        if(b!=0) {
            int res = a / b;
            result.setText(res+"");
        }
        else
            result.setText("Error: Divsion / 0");
    });
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result: "
        android:textSize="18sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"

```

```

        app:layout_constraintHorizontal_bias="0.252"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

<TextView
    android:id="@+id/result"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textSize="18sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.52"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<EditText
    android:id="@+id/no1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="Enter First Number"
    android:inputType="text"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.437"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.228" />

<EditText
    android:id="@+id/no2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="Enter Second Number"
    android:inputType="text"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.437"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.355" />

```

```

<Button
    android:id="@+id/ad"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Add"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.275"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.655" />

```

```

<Button
    android:id="@+id/sub"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Subtract"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.64"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.655" />

```

```

<Button
    android:id="@+id/mult"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Multiply"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.275"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.78" />

```

```

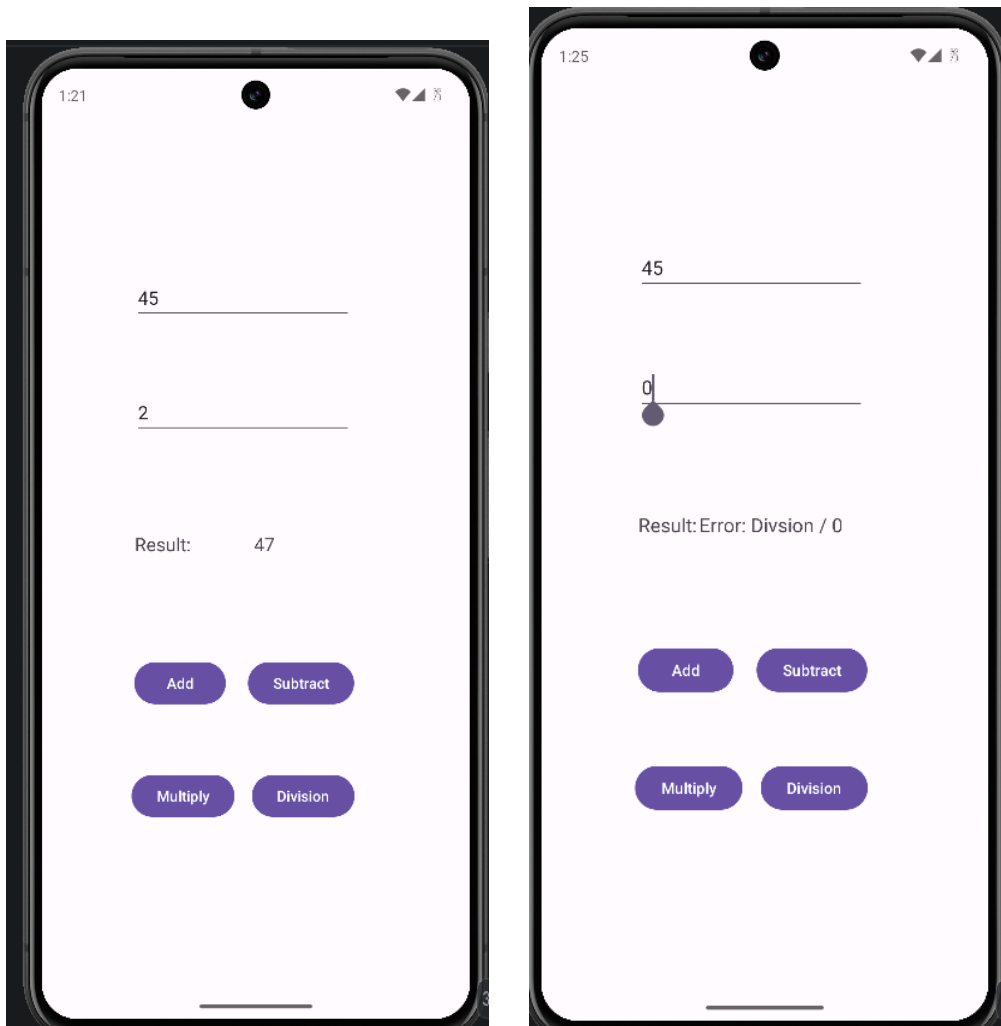
<Button
    android:id="@+id/div"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Division"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"

```

```
app:layout_constraintHorizontal_bias="0.646"  
app:layout_constraintStart_toStartOf="parent"  
app:layout_constraintTop_toTopOf="parent"  
app:layout_constraintVertical_bias="0.78" />
```

</androidx.constraintlayout.widget.ConstraintLayout>

Output:



Program No: 5

Date: 29/06/2026

Program Title: Create an app that changes text color when a button is clicked.

MainActivity.java:

```
package com.example.pg05_textcolor;

import android.graphics.Color;
import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.widget.TextView;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {
    TextView name;
    Button color;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        color = findViewById(R.id.color);
        name = findViewById(R.id.name);
        color.setOnClickListener(v -> {
            name.setTextColor(Color.RED);
        });
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```

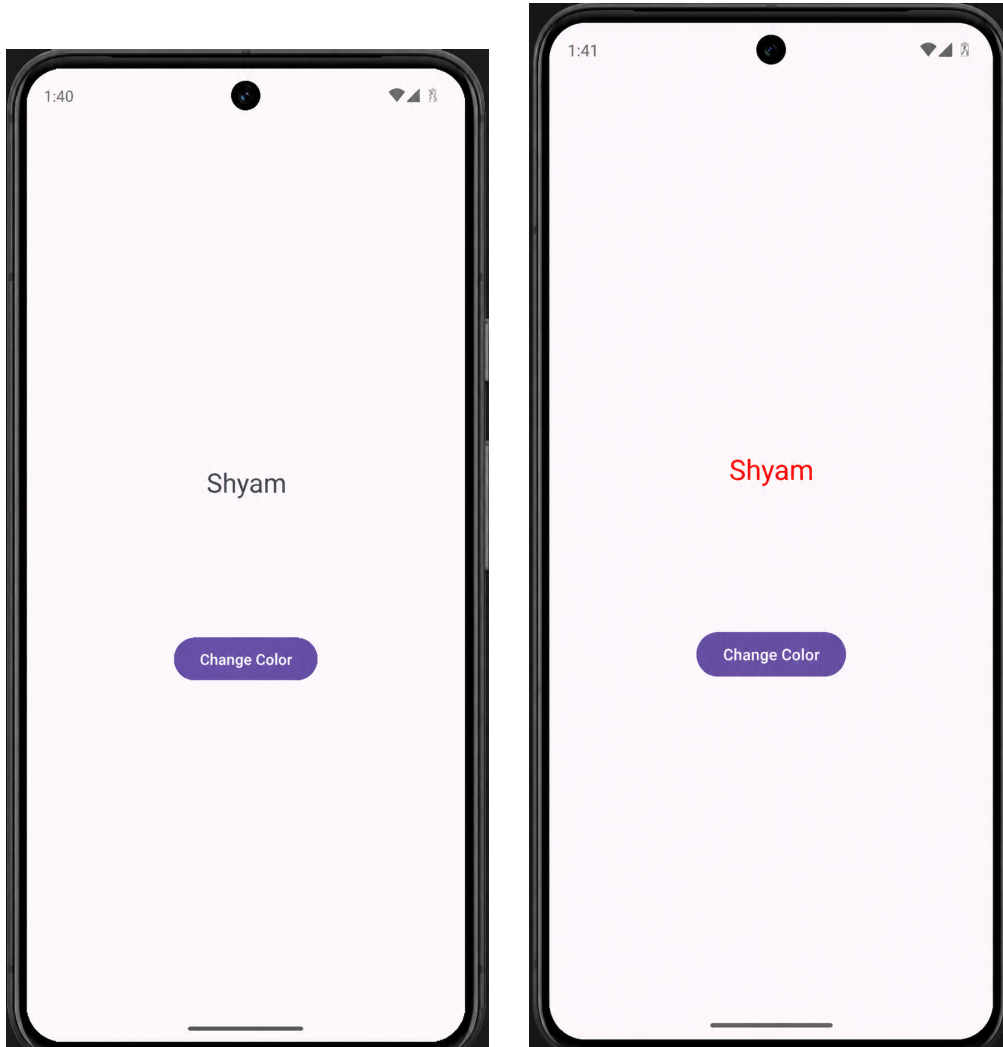
<TextView
    android:id="@+id/name"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Shyam"
    android:textSize="24sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.498"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.421" />

<Button
    android:id="@+id/color"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Change Color"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.613" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program No: 6

Date: 29/06/2026

Program Title: Create an app that displays Date and Time in different formats using Linear Layout.

MainActivity.java:

```
package com.example.pg06_datetime;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.Locale;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        final TextView textView1 = findViewById(R.id.textView1);
        final TextView textView2 = findViewById(R.id.textView2);
        final TextView textView3 = findViewById(R.id.textView3);
        final TextView textView4 = findViewById(R.id.textView4);
        Button button1 = findViewById(R.id.button1);
        button1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Date currentDateTime = new Date();
                SimpleDateFormat format1 = new SimpleDateFormat("EEEE, dd
MMM yyyy", Locale.getDefault());
                SimpleDateFormat format2 = new SimpleDateFormat("MM/dd/yyyy",
Locale.getDefault());
                SimpleDateFormat format3 = new SimpleDateFormat("HH:mm:ss
z", Locale.getDefault());
                SimpleDateFormat format4 = new SimpleDateFormat("hh:mm a",
Locale.getDefault());
                textView1.setText(format1.format(currentDateTime));
                textView2.setText(format2.format(currentDateTime));
                textView3.setText(format3.format(currentDateTime));
                textView4.setText(format4.format(currentDateTime));
            }
        });
    }
}
```



```

    });
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp" />

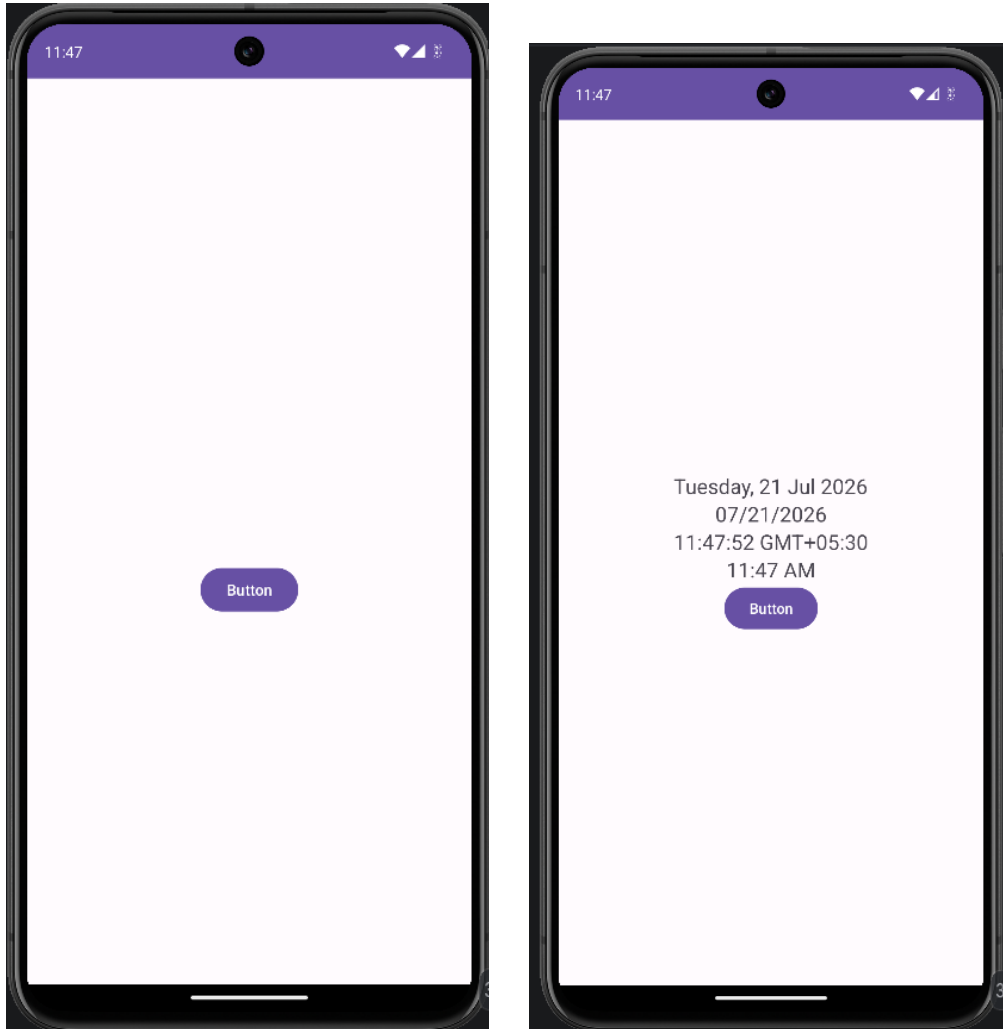
    <TextView
        android:id="@+id/textView4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp" />

    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button" />

```

`</LinearLayout>`

Output:



Program No: 7

Date: 29/06/2026

Program Title: Create an app that displays a Toast on the launch of an activity.

MainActivity.java:

```
package com.example.pg07_launchtoast;

import android.os.Bundle;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        Toast.makeText(this, "Welcome Shyam", Toast.LENGTH_SHORT).show();
    }
}
```

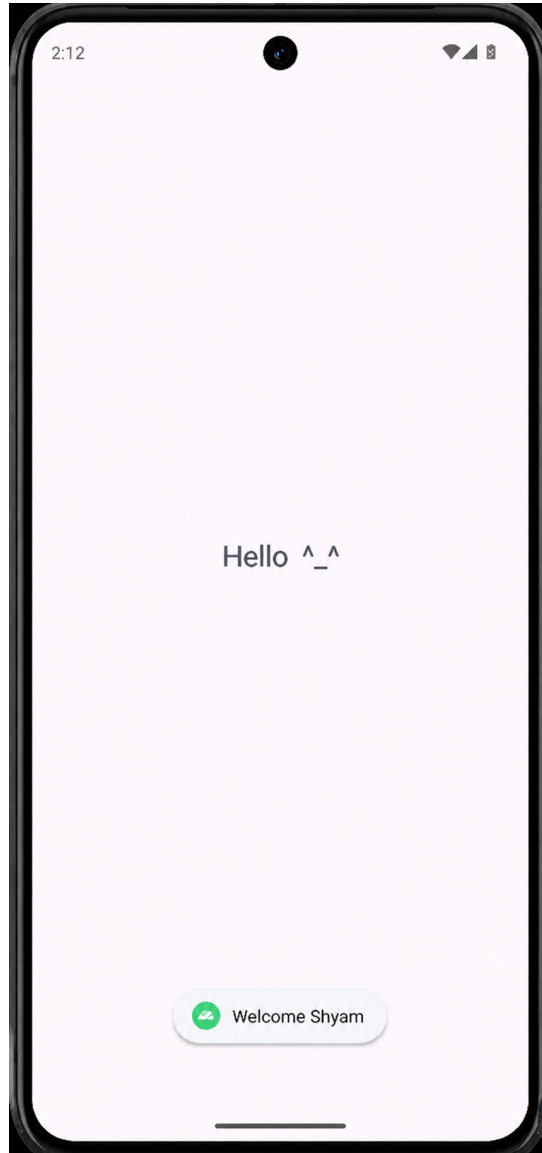
activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello ^_^"
        android:textSize="24sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 8

Date: 29/06/2026

Program Title: Create an app that creates an Alert Dialog Box.

MainActivity.java:

```
package com.example.pg08.alertbox;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AlertDialog;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btn = findViewById(R.id.btn);
        btn.setOnClickListener(v -> {
            AlertDialog.Builder builder = new AlertDialog.Builder(MainActivity.this);
            builder.setTitle("Alert");
            builder.setMessage("What do you want to click?");

            builder.setPositiveButton("Yes", (dialog, which) -> {
                Toast.makeText(MainActivity.this, "You clicked Yes",
                    Toast.LENGTH_SHORT).show();
            });

            builder.setNegativeButton("No", (dialog, which) -> {
                Toast.makeText(MainActivity.this, "You clicked No",
                    Toast.LENGTH_SHORT).show();
            });

            builder.setNeutralButton("Cancel", (dialog, which) -> {
                dialog.dismiss();
            });

            builder.show();
        });
    }
}
```

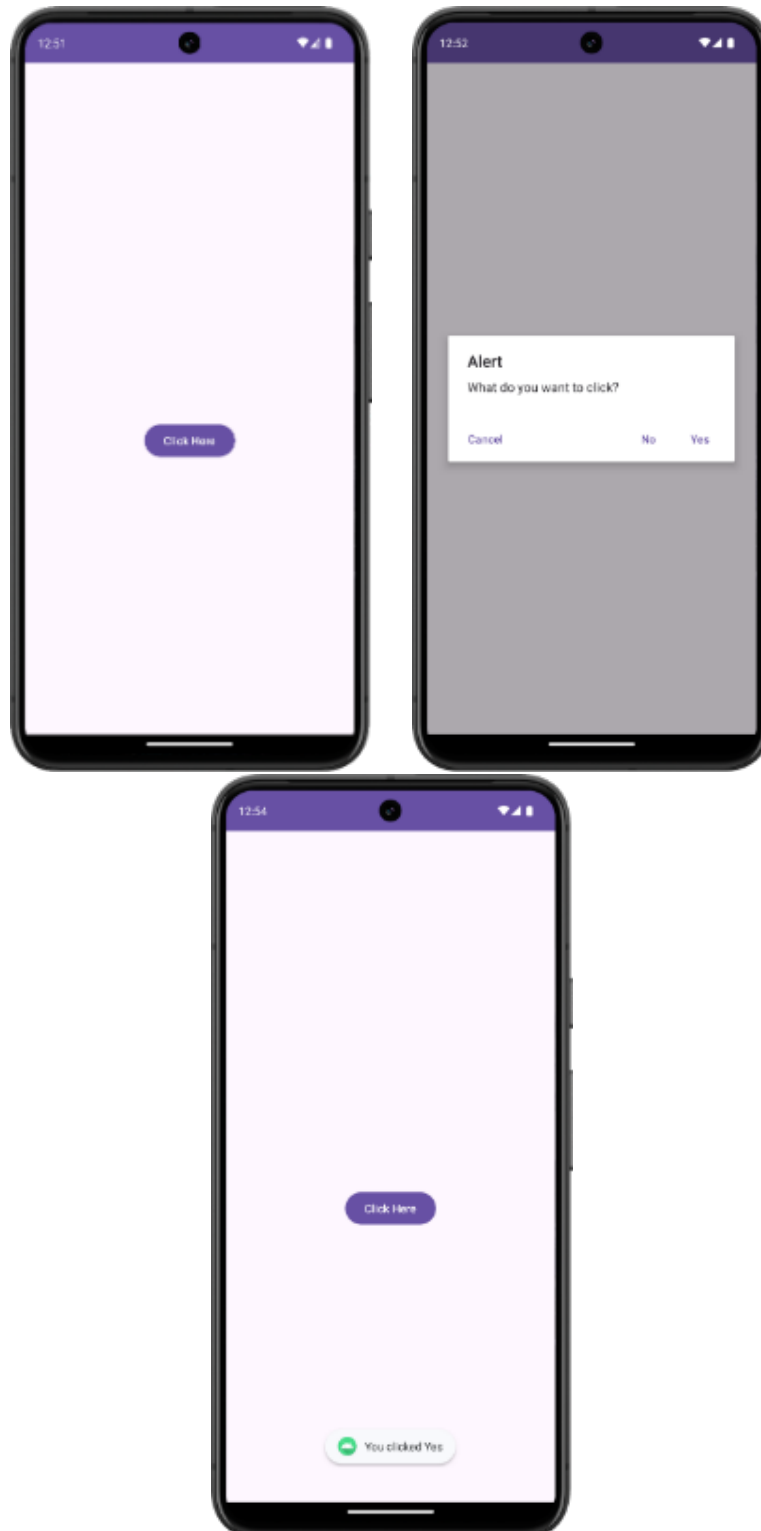
activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click Here"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.566" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 9

Date: 29/06/2026

Program Title: Create a Greeting card app that displays an image.

MainActivity.java:

```
package com.example.pg09_greetingcard;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="288dp"
        android:layout_height="393dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.495"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.328"
        app:srcCompat="@drawable/greetings" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```



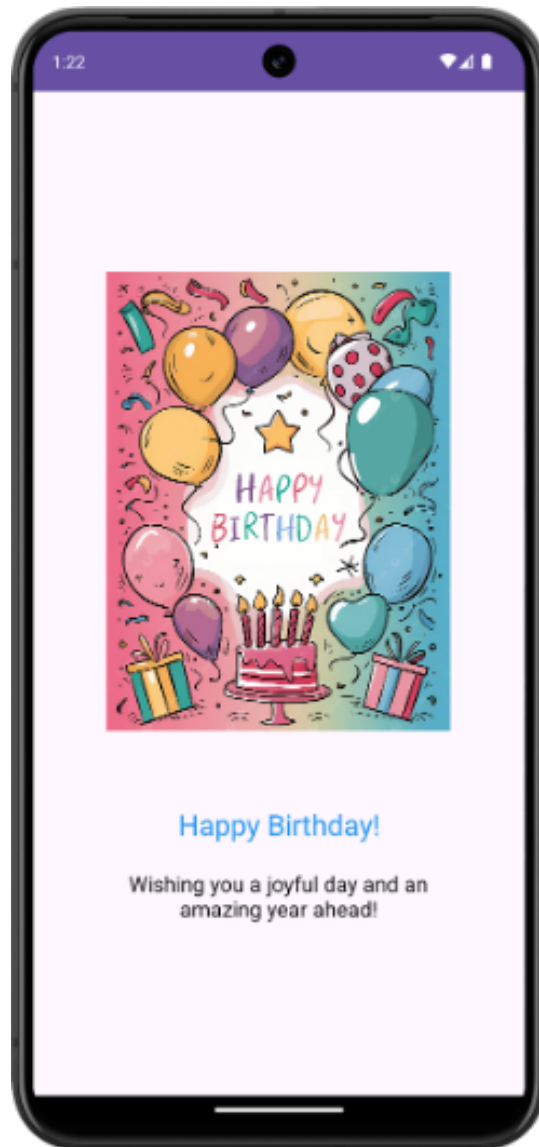
```
    android:text="Happy Birthday!"
    android:textColor="#2196F3"
    android:textSize="24sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.739" />
```

```
<TextView
```

```
    android:id="@+id/textView"
    android:layout_width="258dp"
    android:layout_height="59dp"
    android:text="Wishing you a joyful day and an amazing year ahead!"
    android:textAlignment="center"
    android:textColor="#020202"
    android:textSize="18sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.496"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.833" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 10

Date: 29/06/2026

Program Title: Create an app that plays audio when we click on the button “Play Music” and plays video when we click on the button “Play Video”.

MainActivity.java:

```
package com.example.pg10_videomusic;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.widget.Button;
import android.widget.VideoView;
import android.net.Uri;
import android.media.MediaPlayer;
import android.widget.MediaController;

public class MainActivity extends AppCompatActivity {
    MediaPlayer mediaPlayer;
    VideoView videoView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btnMusic = findViewById(R.id.music);
        Button btnVideo = findViewById(R.id.video);

        videoView = findViewById(R.id.videoView);
        videoView.setZOrderOnTop(true);

        mediaPlayer = MediaPlayer.create(this, R.raw.song);

        btnMusic.setOnClickListener(v -> {
            if (!mediaPlayer.isPlaying()) {
                mediaPlayer.start();
            }
        });

        btnVideo.setOnClickListener(v -> {
            Uri uri = Uri.parse("android.resource://" + getPackageName()
                + "/" + R.raw.video);
            videoView.setVideoURI(uri);
            MediaController controller =
```

```

        new MediaController(MainActivity.this);
        videoView.setMediaController(controller);
        controller.setAnchorView(videoView);
        videoView.start();
    });
}

@Override
protected void onDestroy() {
    super.onDestroy();
    if (mediaPlayer != null) {
        mediaPlayer.release();
        mediaPlayer = null;
    }
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

    <VideoView
        android:id="@+id/videoView"
        android:layout_width="348dp"
        android:layout_height="231dp"
        android:background="#000000"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.492"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.23" />

    <Button
        android:id="@+id/music"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"

```

```

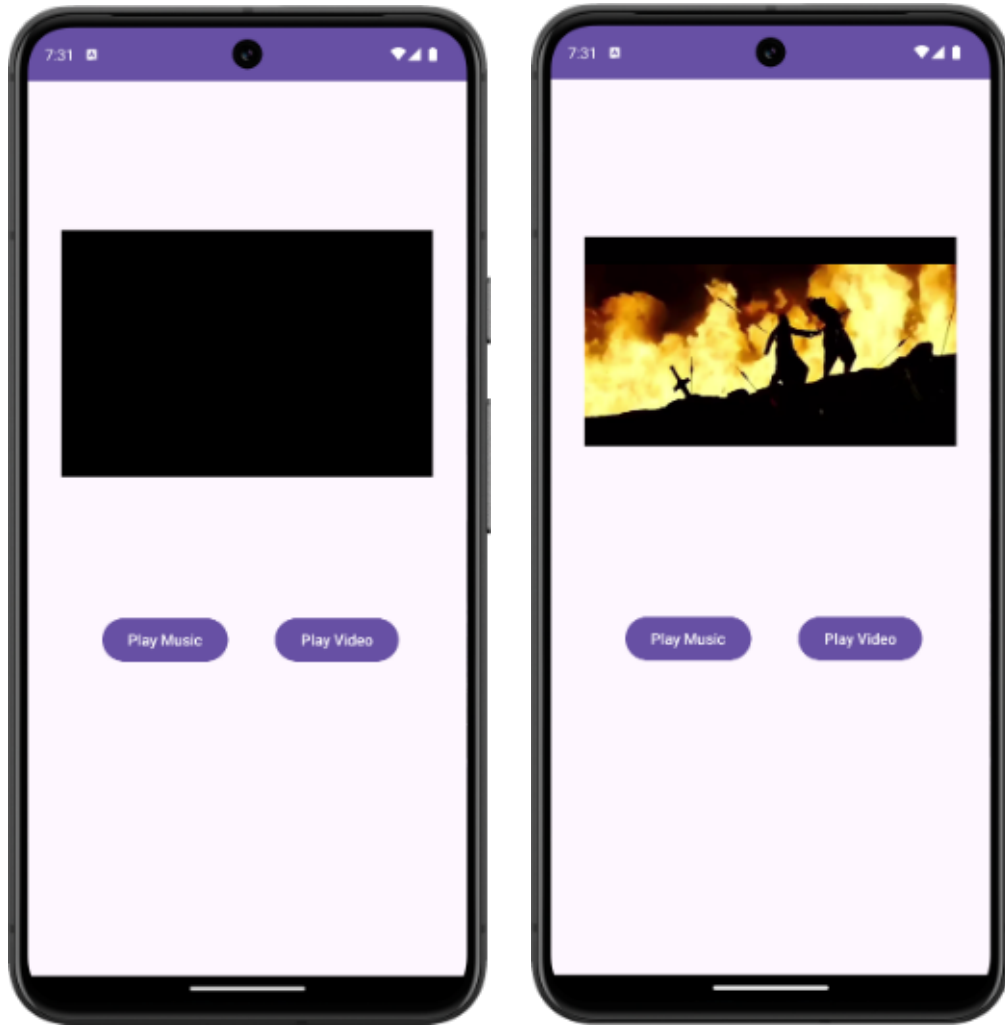
        android:text="Play Music"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.237"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.632" />

<Button
    android:id="@+id/video"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="8dp"
    android:text="Play Video"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.777"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.632" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program No: 11

Date: 29/06/2026

Program Title: Create an app that plays video from the internet.

MainActivity.java:

```
package com.example.pg11_internetvideo;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.widget.Button;
import android.widget.VideoView;
import android.net.Uri;
import android.widget.MediaController;

public class MainActivity extends AppCompatActivity {
    VideoView videoView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btnVideo = findViewById(R.id.play);
        videoView = findViewById(R.id.videoView);
        videoView.setZOrderOnTop(true);

        btnVideo.setOnClickListener(v -> {
            Uri uri = Uri.parse("https://www.learningcontainer.com/"
                                + "wp-content/uploads/2020/05/"
                                + "sample-mp4-file.mp4");
            videoView.setVideoURI(uri);
            MediaController controller =
                new MediaController(MainActivity.this);
            videoView.setMediaController(controller);
            controller.setAnchorView(videoView);
            videoView.start();
        });
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
```

```

xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

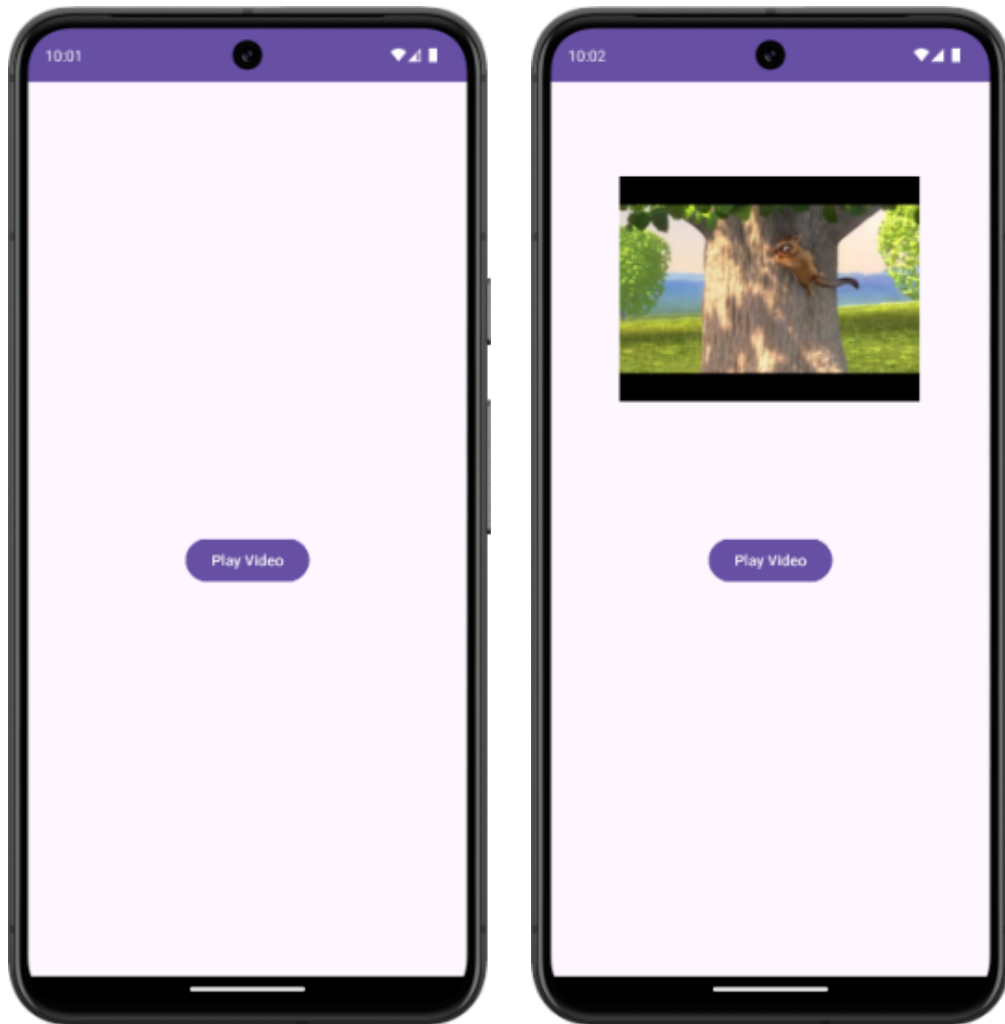
<VideoView
    android:id="@+id/videoView"
    android:layout_width="365dp"
    android:layout_height="211dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.493"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.141" />

<Button
    android:id="@+id/play"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Play Video"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.537" />

</androidx.constraintlayout.widget.ConstraintLayout>

```


Output:



Program No: 12

Date: 29/06/2026

Program Title: Create an app to demonstrate Options Menu.

MainActivity.java:

```
package com.example.pg12.optionsmenu;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int id = item.getItemId();
        if (id == R.id.settings) {
            Toast.makeText(this, "Settings Selected",
                           Toast.LENGTH_SHORT).show();
        }
        else if (id == R.id.about) {
            Toast.makeText(this, "About Selected",
                           Toast.LENGTH_SHORT).show();
        }
        else if (id == R.id.exit) {
            finish();
        }
        return true;
    }
}
```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"/>

```

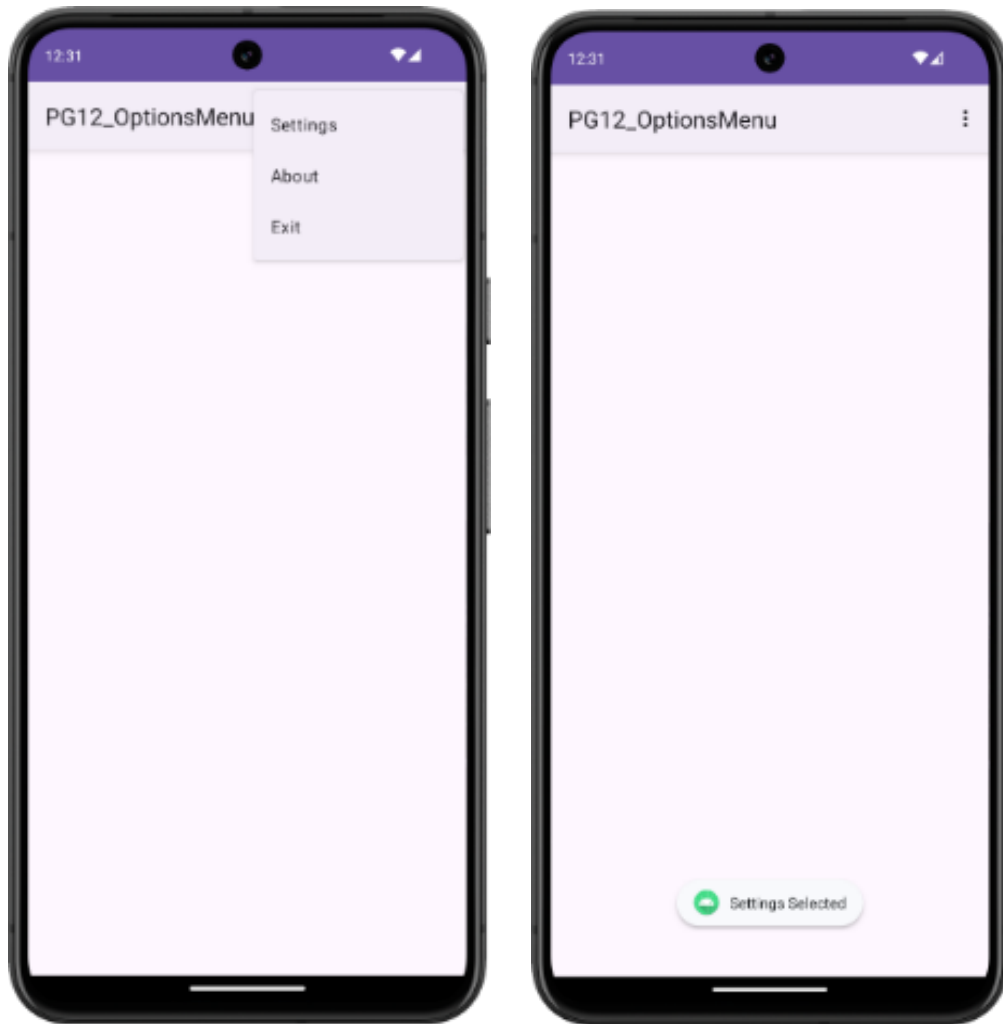
menu.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/settings"
        android:title="Settings" />
    <item
        android:id="@+id/about"
        android:title="About" />
    <item
        android:id="@+id/exit"
        android:title="Exit" />
</menu>

```

Output:



Program No: 13

Date: 29/06/2026

Program Title: Create an app that changes the background color of an activity using Context Menu.

MainActivity.java:

```
package com.example.context_menu;

import android.graphics.Color;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.view.ContextMenu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.view.View;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
    View layout;
    TextView tx1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        layout = findViewById(R.id.main);
        tx1 = findViewById(R.id.tx1);
        registerForContextMenu(tx1);
    }

    @Override
    public void onCreateContextMenu(ContextMenu menu, View v,
                                   ContextMenu.ContextMenuInfo menuinfo){
        super.onCreateContextMenu(menu, v, menuinfo);
        MenuInflater inflater = getMenuInflater();
        inflater.inflate(R.menu.menu_main, menu);
    }

    public boolean onContextItemSelected(MenuItem item) {
        if (item.getItemId() == R.id.red){
            layout.setBackgroundColor(Color.RED);
        }
        if (item.getItemId() == R.id.blue) {
            layout.setBackgroundColor(Color.BLUE);
        }
    }
}
```

```

    }
    if (item.getItemId() == R.id.green) {
        layout.setBackgroundColor(Color.GREEN);
    }
    if (item.getItemId() == R.id.yellow) {
        layout.setBackgroundColor(Color.YELLOW);
    }
    return false;
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/tx1"
        android:layout_width="90dp"
        android:layout_height="30dp"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.454"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.499" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

menu.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/red"
        android:title="RED" />
    <item
        android:id="@+id/blue"

```

```
        android:title="BLUE" />
    <item
        android:id="@+id/green"
        android:title="GREEN" />
    <item
        android:id="@+id/yellow"
        android:title="YELLOW" />
</menu>
```

Output:



Program No: 14

Date: 29/06/2026

Program Title: Create an app to demonstrate how radiobuttons and checkboxes can be used in android.

MainActivity.java:

```
package com.example.radiocheckboxbox;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.widget.Button;
import android.widget.RadioGroup;
import android.widget.RadioButton;
import android.widget.CheckBox;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    RadioGroup gender;
    CheckBox ten, twelve, UG, PG;
    Button submit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        gender = findViewById(R.id.gender);
        ten = findViewById(R.id.ten);
        twelve = findViewById(R.id.twelve);
        UG = findViewById(R.id.UG);
        PG = findViewById(R.id.PG);
        submit = findViewById(R.id.submit);

        submit.setOnClickListener(v ->{
            int selectedId = gender.getCheckedRadioButtonId();
            RadioButton rb = findViewById(selectedId);
            String gen = rb.getText().toString();
            StringBuilder education = new StringBuilder();

            if (ten.isChecked()) education.append("Tenth ");
            if (twelve.isChecked()) education.append("Plus Two ");
            if (UG.isChecked()) education.append("Undergraduate ");
            if (PG.isChecked()) education.append("Postgraduate ");
```



```

        Toast.makeText(this, gen + " | " + education,
                        Toast.LENGTH_LONG).show();
    });
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#FB9E9E"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView4"
        android:layout_width="82dp"
        android:layout_height="23dp"
        android:text="Gender: "
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.124"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.114" />

    <RadioGroup
        android:id="@+id/gender"
        android:layout_width="247dp"
        android:layout_height="143dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.25"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.205">
        <RadioButton
            android:id="@+id/male"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"

```

```

        android:text="Male" />
<RadioButton
    android:id="@+id/female"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Female" />
<RadioButton
    android:id="@+id/tg"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="TG" />
</RadioGroup>

<CheckBox
    android:id="@+id/ten"
    android:layout_width="134dp"
    android:layout_height="43dp"
    android:text="X"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.155"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.527" />

<CheckBox
    android:id="@+id/UG"
    android:layout_width="133dp"
    android:layout_height="48dp"
    android:text="UG"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.154"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.71" />

<CheckBox
    android:id="@+id/twelve"
    android:layout_width="134dp"
    android:layout_height="48dp"
    android:text="XII"
    app:layout_constraintBottom_toBottomOf="parent"

```

```

        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.158"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.619" />

<Button
    android:id="@+id/submit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Submit"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.468"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.915" />

<TextView
    android:id="@+id/textView"
    android:layout_width="90dp"
    android:layout_height="22dp"
    android:text="Qualification"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.13"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.452" />

<CheckBox
    android:id="@+id/PG"
    android:layout_width="133dp"
    android:layout_height="48dp"
    android:text="PG"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.154"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.808" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:

The image displays two side-by-side smartphone screens showing a form interface. The left screen shows the initial state where all options are unselected. The right screen shows the state after selection, with a confirmation toast message at the bottom.

Left Screen (12:50):

- Gender:
 - ☐ Male
 - ☐ Female
 - ☐ TG
- Qualification
 - ☐ X
 - ☐ XII
 - ☐ UG
 - ☐ PG
-

Right Screen (12:53):

- Gender:
 - ☒ Male
 - ☐ Female
 - ☐ TG
- Qualification
 - ☒ X
 - ☒ XII
 - ☒ UG
 - ☐ PG
- (partially obscured)

Toast Message: Male | Tenth Plus Two Undergraduate

Program No: 15

Date: 29/06/2026

Program Title: Create an app that launches a new activity on a button click

MainActivity.java:

```
package com.example.a2screen_welcome;
import android.content.Intent;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button btn = findViewById(R.id.button);
        btn.setOnClickListener(v-> {
            Intent intent = new Intent(MainActivity.this,
                                     SecondaryActivity.class);
            startActivity(intent);
        });
    }
}
```

SecondaryActivity.java:

```
package com.example.a2screen_welcome;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class SecondaryActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_secondary);
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
```

```

        android:layout_height="match_parent"
        tools:context=".MainActivity">

        <Button
            android:id="@+id/button"
            android:layout_width="172dp"
            android:layout_height="74dp"
            android:text="Go to Next Page"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintHorizontal_bias="0.498"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toTopOf="parent"
            app:layout_constraintVertical_bias="0.558" />
    </androidx.constraintlayout.widget.ConstraintLayout>

```

activity_secondary.xml:

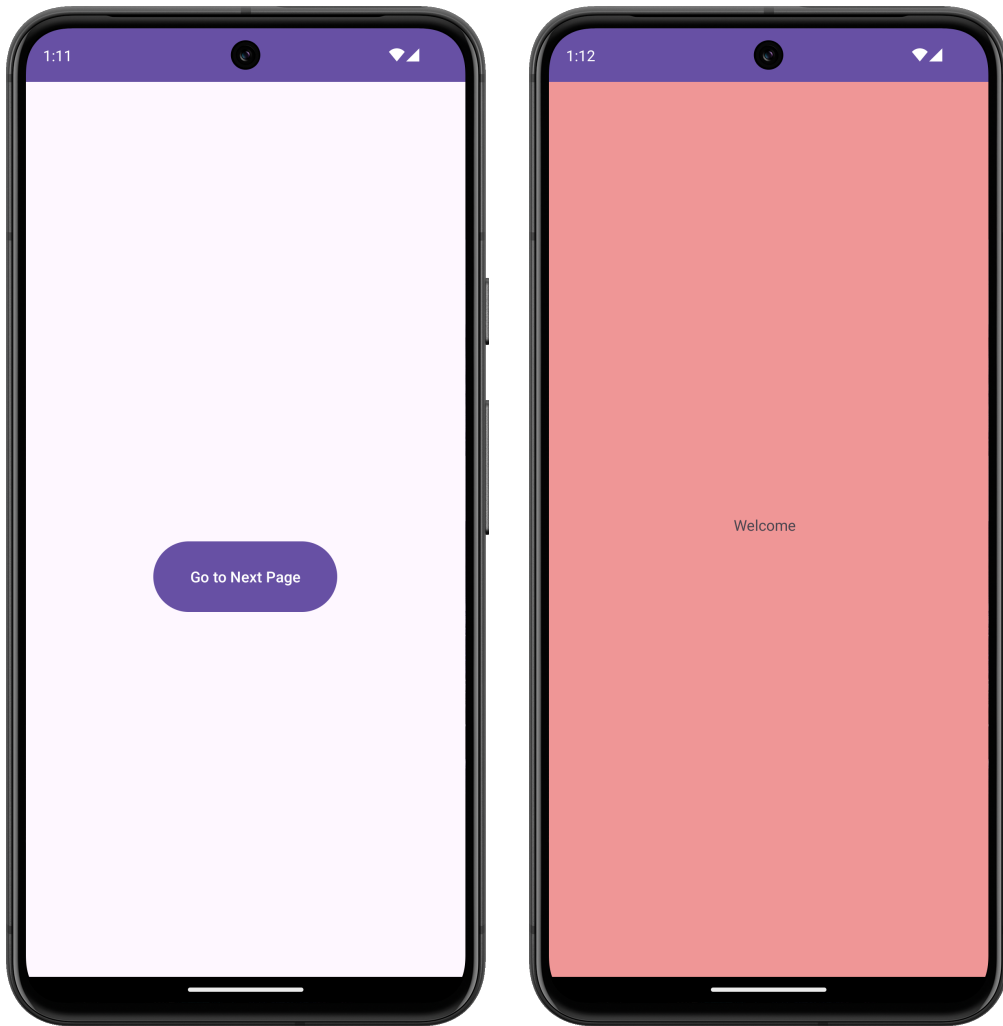
```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#EF9696"
    tools:context=".SecondaryActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="66dp"
        android:layout_height="28dp"
        android:text="Welcome"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program No: 16

Date: 29/06/2026

Program Title: Create an app that launches a new activity on a button click to display student details with checkbox and radio buttons.

MainActivity.java:

```
package com.example.pg16_studentdetails;

import android.content.Intent;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button submit = findViewById(R.id.submit);
        EditText name = findViewById(R.id.name);
        EditText mobile = findViewById(R.id.mobile);
        RadioGroup department = findViewById(R.id.department);
        RadioGroup gender = findViewById(R.id.gender);
        CheckBox ai = findViewById(R.id.ai);
        CheckBox cloud = findViewById(R.id.cloud);
        CheckBox auto = findViewById(R.id.autocad);
        CheckBox ship = findViewById(R.id.ship);

        submit.setOnClickListener(v -> {
            int depId = department.getCheckedRadioButtonId();
            RadioButton rdep = findViewById(depId);
            int genId = gender.getCheckedRadioButtonId();
            RadioButton rgen = findViewById(genId);
            StringBuilder sb = new StringBuilder();

            if (ai.isChecked())
                sb.append("AI/ML\n");
```



```

        if (cloud.isChecked())
            sb.append("Cloud Computing\n");

        if (auto.isChecked())
            sb.append("AutoCAD\n");

        if (ship.isChecked())
            sb.append("Ship Technology\n");

        Intent intent = new Intent(MainActivity.this, Details.class);
        intent.putExtra("name", name.getText().toString());
        intent.putExtra("mobile", mobile.getText().toString());
        intent.putExtra("courses", sb.toString());
        intent.putExtra("department", rdep.getText().toString());
        intent.putExtra("gender", rgen.getText().toString());
        startActivity(intent);
    });
}
}

```

SecondaryActivity.java:

```

package com.example.pg16_studentdetails;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.widget.TextView;

public class Details extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_details);

        String name = getIntent().getStringExtra("name");
        String mobile = getIntent().getStringExtra("mobile");
        String department = getIntent().getStringExtra("department");
        String gender = getIntent().getStringExtra("gender");
        String courses = getIntent().getStringExtra("courses");

        TextView details = findViewById(R.id.details);
        details.setText("Name: " + name +
            "\nMobile: " + mobile +
            "\nDepartment: " + department +

```

```

        "\nGender: " + gender +
        "\nCourses:\n" + courses);
    }
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="68dp"
        android:layout_height="24dp"
        android:text="Name: "
        android:textSize="16sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.11"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.077" />

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Department: "
        android:textSize="16sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.119"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.131" />

    <RadioGroup
        android:id="@+id/department"
        android:layout_width="314dp"

```

```

        android:layout_height="134dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.391"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.207">

        <RadioButton
            android:id="@+id/cs"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Computer Science"
            android:textSize="16sp" />

        <RadioButton
            android:id="@+id/civil"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Civil Engineering"
            android:textSize="16sp" />

        <RadioButton
            android:id="@+id/mech"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Mechanical Enginneering"
            android:textSize="16sp" />
    </RadioGroup>

    <EditText
        android:id="@+id/name"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:hint="Name"
        android:inputType="text"
        android:textSize="16sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.719"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"

```

```

        app:layout_constraintVertical.bias="0.076" />

<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Preferred Course: "
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal.bias="0.123"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical.bias="0.385" />

<CheckBox
    android:id="@+id/ai"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="AI/ML"
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal.bias="0.133"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical.bias="0.44" />

<CheckBox
    android:id="@+id/ship"
    android:layout_width="127dp"
    android:layout_height="41dp"
    android:text="Ship Technology"
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal.bias="0.133"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical.bias="0.517" />

<CheckBox
    android:id="@+id/cloud"

```

```

        android:layout_width="127dp"
        android:layout_height="41dp"
        android:text="Cloud Computing"
        android:textSize="16sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.693"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.44" />

```

<CheckBox

```

        android:id="@+id/autocad"
        android:layout_width="127dp"
        android:layout_height="41dp"
        android:text="Autocad"
        android:textSize="16sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.693"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.517" />

```

<TextView

```

        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Gender: "
        android:textSize="16sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.107"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.585" />

```

<RadioGroup

```

        android:id="@+id/gender"
        android:layout_width="318dp"
        android:layout_height="162dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"

```

```

app:layout_constraintHorizontal_bias="0.408"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.775">

<RadioButton
    android:id="@+id/male"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Male"
    android:textSize="16sp" />

<RadioButton
    android:id="@+id/female"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Female"
    android:textSize="16sp" />

<RadioButton
    android:id="@+id/tg"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="TG"
    android:textSize="16sp" />
</RadioGroup>

<TextView
    android:id="@+id/textView5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Mobile No: "
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.105"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.861" />

<EditText
    android:id="@+id/mobile"
    android:layout_width="wrap_content"

```

```

        android:layout_height="wrap_content"
        android:ems="10"
        android:hint="Mobile No."
        android:inputType="number"
        android:textSize="16sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.674"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.876" />

<Button
    android:id="@+id/submit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Submit"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.976" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

activity_secondary.xml:

```

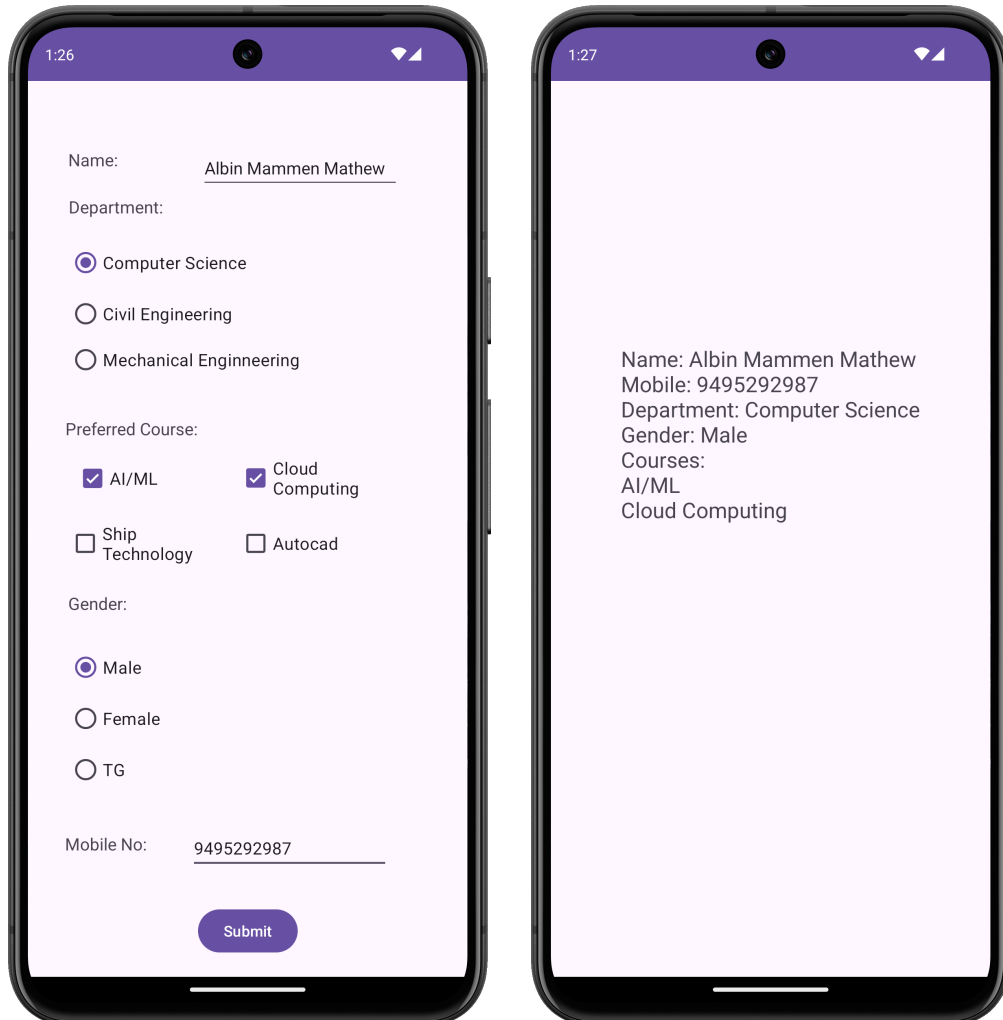
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Details">

    <TextView
        android:id="@+id/details"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"

```

```
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.381" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 17

Date: 29/06/2026

Program Title: Create an app that launches a new activity on a button click to calculate and display employee details and salary.

MainActivity.java:

```
package com.example.pg17_employeeedetails;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText name, id, sal;
    Button btnNext;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        name = findViewById(R.id.name);
        id = findViewById(R.id.id);
        sal = findViewById(R.id.sal);
        btnNext = findViewById(R.id.btnNext);

        btnNext.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent = new Intent(MainActivity.this,
                                           SecondaryActivity.class);
                intent.putExtra("name", name.getText().toString());
                intent.putExtra("id", id.getText().toString());
                intent.putExtra("salary", sal.getText().toString());
                startActivity(intent);
            }
        });
    }
}
```

SecondaryActivity.java:

```
package com.example.pg17_employeeedetails;
```

```

import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class SecondaryActivity extends AppCompatActivity {
    TextView result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_secondary);

        result = findViewById(R.id.result);
        Intent intent = getIntent();

        if (intent != null) {
            String name = intent.getStringExtra("name");
            String id = intent.getStringExtra("id");
            String salaryStr = intent.getStringExtra("salary");

            double basic = 0;
            try {
                if (salaryStr != null) {
                    basic = Double.parseDouble(salaryStr);
                }
            } catch (Exception e) {
                // handle parsing error
            }

            double hra = basic * 0.15;
            double da = basic * 0.10;
            double gross = basic + hra + da;

            result.setText(
                "Employee Details\n\n" +
                "Name : " + name +
                "\nEmployee ID : " + id +
                "\nBasic Salary : " + basic +
                "\nHRA (15%) : " + hra +
                "\nDA (10%) : " + da +
                "\nGross Salary : " + gross
            );
        }
    }
}

```

```

    );
}
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
android:layout_width="match_parent"
android:layout_height="match_parent">

<TextView
    android:id="@+id/tvTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Employee Details"
    android:textSize="22sp"
    android:textStyle="bold"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.587"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.067" />

<TextView
    android:id="@+id/tvName"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="20dp"
    android:layout_marginTop="20dp"
    android:text="Employee Name"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@id/tvTitle" />

<EditText
    android:id="@+id/name"
    android:layout_width="250dp"
    android:layout_height="wrap_content"
    android:hint="Enter Name"
    app:layout_constraintStart_toStartOf="@id/tvName"
    app:layout_constraintTop_toBottomOf="@id/tvName" />

```

```

<TextView
    android:id="@+id/tvId"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="15dp"
    android:text="Employee ID"
    app:layout_constraintStart_toStartOf="@id/tvName"
    app:layout_constraintTop_toBottomOf="@id/name" />

<EditText
    android:id="@+id/id"
    android:layout_width="250dp"
    android:layout_height="wrap_content"
    android:hint="Enter ID"
    app:layout_constraintStart_toStartOf="@id/tvName"
    app:layout_constraintTop_toBottomOf="@id/tvId" />

<TextView
    android:id="@+id/tvSalary"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="15dp"
    android:text="Basic Salary"
    app:layout_constraintStart_toStartOf="@id/tvName"
    app:layout_constraintTop_toBottomOf="@id/id" />

<EditText
    android:id="@+id/sal"
    android:layout_width="250dp"
    android:layout_height="wrap_content"
    android:hint="Enter Salary"
    android:inputType="numberDecimal"
    app:layout_constraintStart_toStartOf="@id/tvName"
    app:layout_constraintTop_toBottomOf="@id/tvSalary" />

<Button
    android:id="@+id/btnNext"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="25dp"
    android:text="Show Details"
    app:layout_constraintEnd_toEndOf="parent"

```

```
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/sal" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

activity_secondary.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/result"
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:textSize="18sp"
        android:textStyle="bold"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:

The image displays two smartphone screens side-by-side, illustrating the user interface of an application. The left screen shows the input form, and the right screen shows the calculated output.

Left Screen (Input Form):

- Time: 2:16
- Title: Employee Details
- Employee Name: Enter Name
- Employee ID: Enter ID
- Basic Salary: Enter Salary
- Button: Show Details

Right Screen (Output):

- Time: 2:22
- Title: Employee Details
- Name : Albin
- Employee ID : 404
- Basic Salary : 50000.0
- HRA (15%) : 7500.0
- DA (10%) : 5000.0
- Gross Salary : 62500.0

Program No: 18

Date: 29/06/2026

Program Title: Create an app to broadcast a custom intent.

MainActivity.java:

```
package com.example.broadcast;

import android.content.Intent;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    public static final String MY_ACTION =
        "com.example.broadcast.MY_ACTION";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button send = findViewById(R.id.send);
        send.setOnClickListener(v -> {
            Intent intent = new Intent(MY_ACTION);
            intent.putExtra("message", "Hello from MainActivity!!");
            intent.setPackage(getPackageName());
            sendBroadcast(intent);
        });
    }
}
```

MyReceiver.java:

```
package com.example.broadcast;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.widget.Toast;

public class MyReceiver extends BroadcastReceiver {

    @Override
    public void onReceive(Context context, Intent intent) {
        String message = intent.getStringExtra("message");
        Toast.makeText(context, "Custom Broadcast Received: " + message,
```

```

        Toast.LENGTH_SHORT).show();
    }
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/send"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.52"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.614" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

AndroidManifest.xml:

```

<receiver
    android:name=".MyReceiver"
    android:enabled="true"
    android:exported="true">
    <intent-filter>
        <action android:name="com.example.broadcast.MY_ACTION" />
    </intent-filter>
</receiver>

```


Output:



Program No: 19

Date: 29/06/2026

Program Title: Create an app to broadcast a system intent (AIRPLANE MODE CHANGED).

MainActivity.java:

```
package com.example.pg19_airplanemode;

import android.content.Intent;
import android.content.IntentFilter;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    private final AirplaneModeReceiver receiver = new
    AirplaneModeReceiver();

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        IntentFilter filter =
            new IntentFilter(Intent.ACTION_AIRPLANE_MODE_CHANGED);
        registerReceiver(receiver, filter);
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        unregisterReceiver(receiver);
    }
}
```

AirplaneModeReceiver.java:

```
package com.example.pg19_airplanemode;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.widget.Toast;

public class AirplaneModeReceiver extends BroadcastReceiver {

    @Override
    public void onReceive(Context context, Intent intent) {
```

```

        if (intent.ACTION_AIRPLANE_MODE_CHANGED.equals(
            intent.getAction())) {
            boolean isOn = intent.getBooleanExtra("state", false);
            String msg = isOn ? "Airplane Mode turned ON"
                : "Airplane Mode turned OFF";
            Toast.makeText(context, msg, Toast.LENGTH_SHORT).show();
        }
    }
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity" />

```

AndroidManifest.xml:

```

<receiver
    android:name=".AirplaneModeReceiver"
    android:enabled="true"
    android:exported="true">
</receiver>

```

Output:

